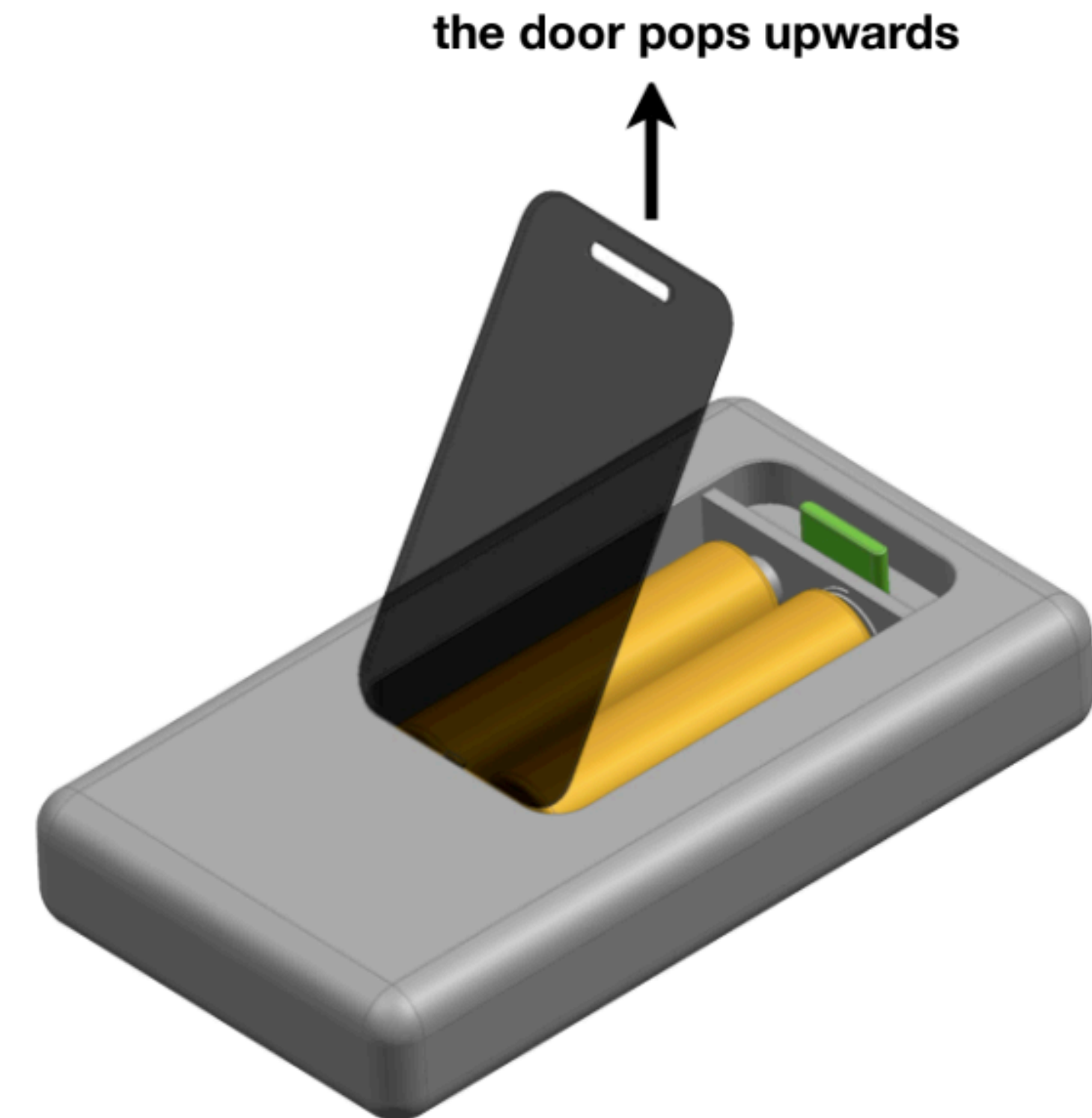
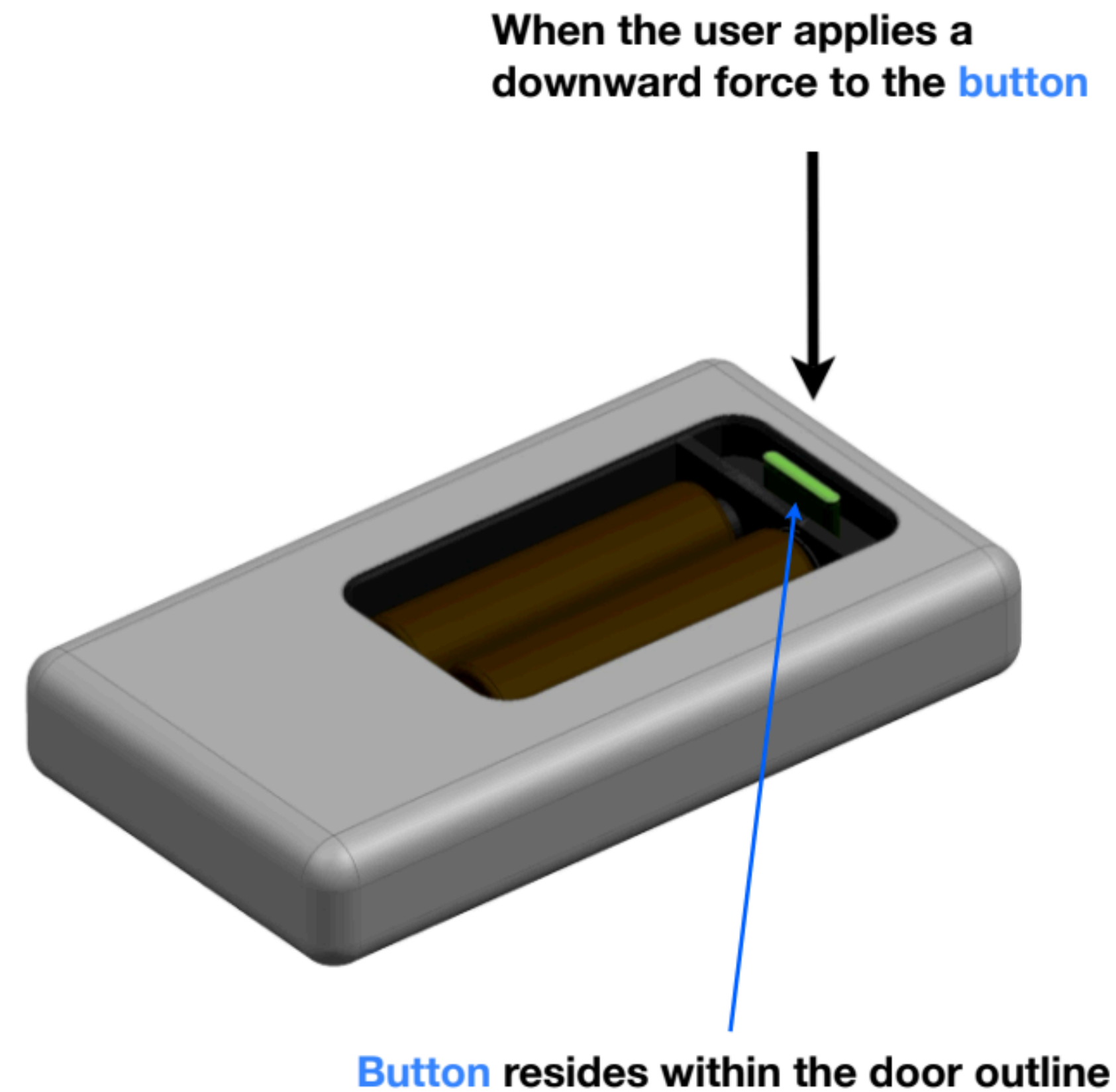




Battery Door Design Challenge

Megan Ying

Assumptions and Requirements

Assumptions

- 5th generation iPod Classic
 - Dimensions: 103.5 x 61.8 x 10.5 mm
 - Mass: 140 g
- Other internal components are out-of-scope
- Users will accept a thicker product
 - AA battery diameter: 14 mm
- Product is cycled < 1000 times over lifespan
 - Duracell AA batteries last ~100 hours during constant use
 - iPod Classic lasts ~5 years
- Product will be mass manufactured (>100,000 units per year)
- Door will pop open when released due to spring-loaded hinge

Requirements

- Button requires minimal user effort to actuate, but resists accidental actuation
 - Button press force of 2-5 N and travel distance of 1-3 mm
- Door require minimal user effort to close, but resists accidental opening
 - Closing force of < 8 N and opening force of > 20 N
- Does not obstruct battery removal or handling
- Prevents rattling or unwanted noise during use
- Endures repeated use over product's lifespan
- Remains functional and secure during drops
- Remains functional under extreme temperatures
- Minimizes production costs and assembly time
- Minimizes additional weight and space footprint
- Maintains sleek and aesthetically pleasing appearance

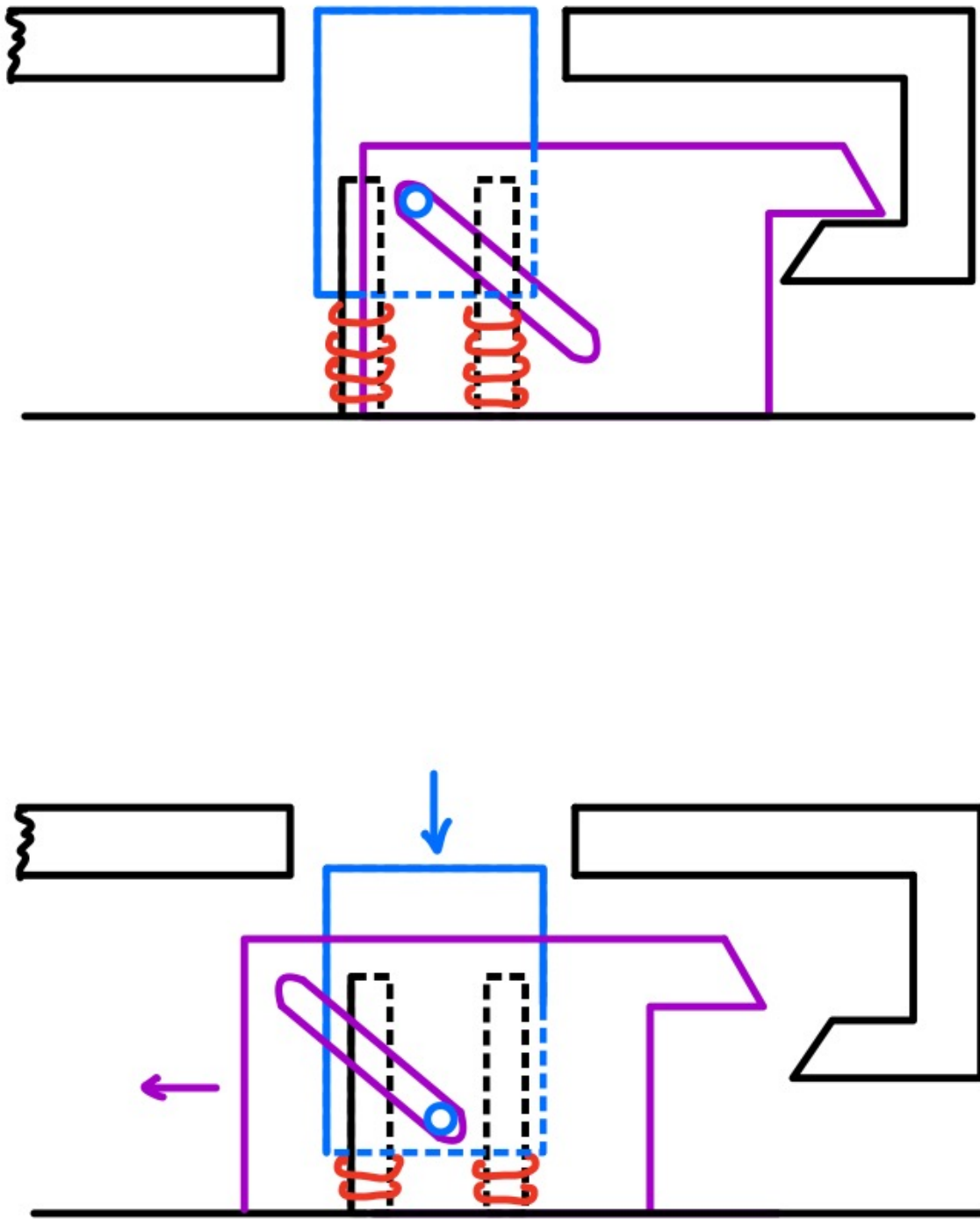
Moving Snap Fit Mechanism

Inspiration



Battery covers and water bottles with snap fits

Concept



1. Latch hooks onto door in locked position

2. Button is pressed and latch moves to release door

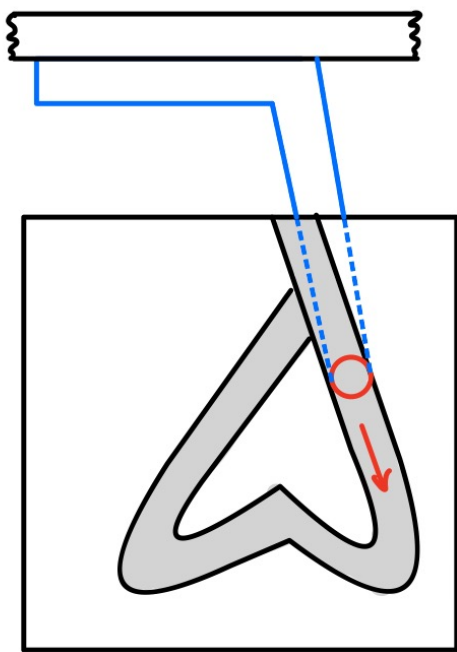
Push Release Mechanism

Inspiration

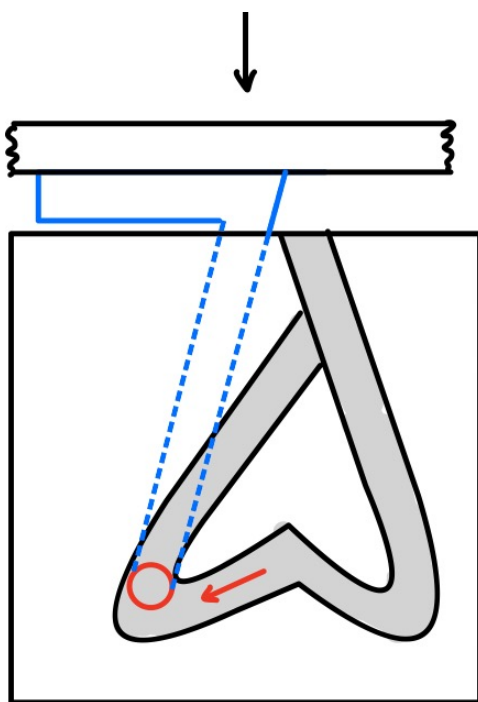


Push to open cabinets with touch latches

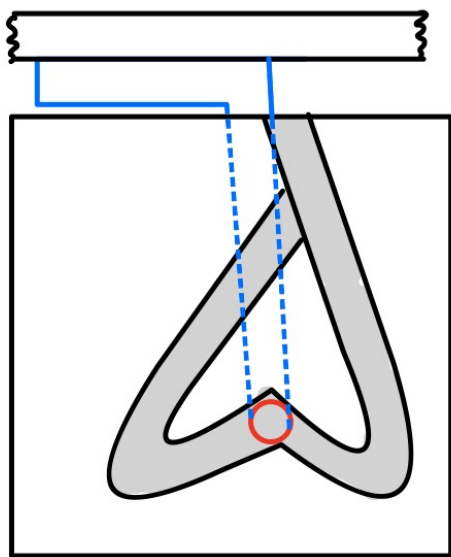
Concept



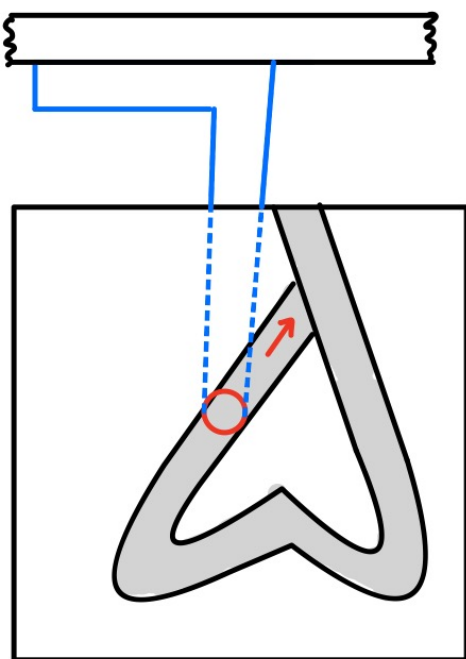
1. Button is pressed and hook travels down groove



3. Button is pressed and hook travels down groove



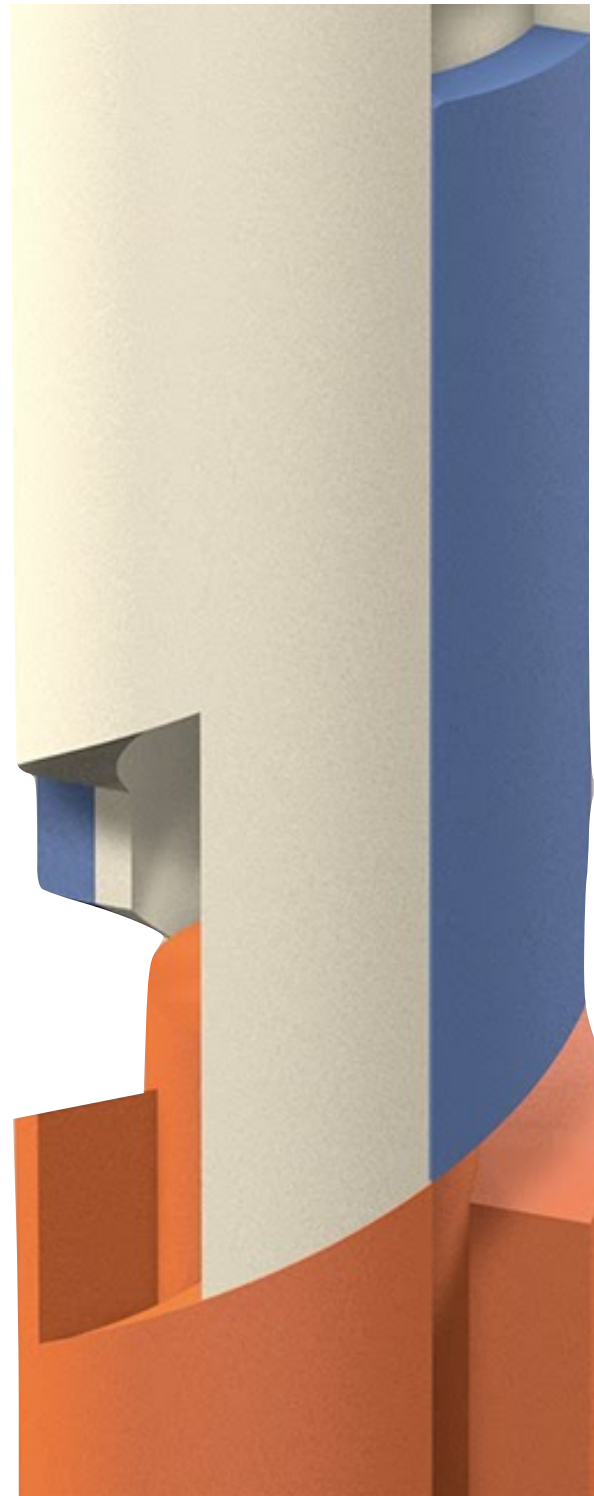
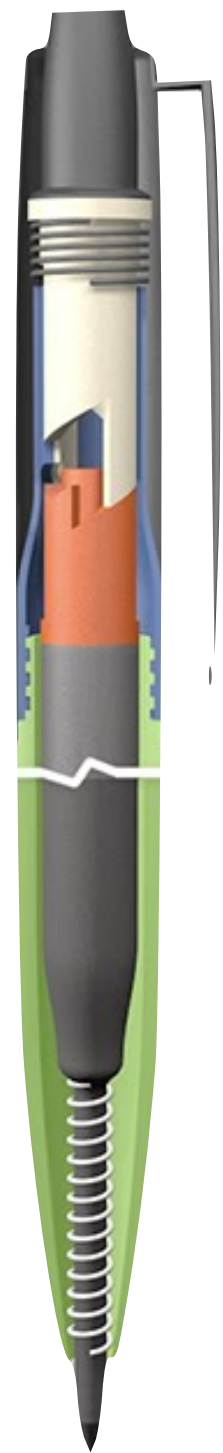
2. Button is released and hook sits in locked position



4. Button is pressed and hook travels to unlocked position

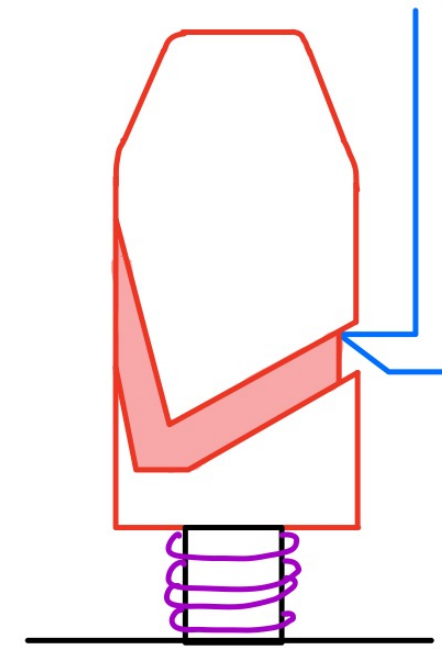
Rotating Body Mechanism

Inspiration

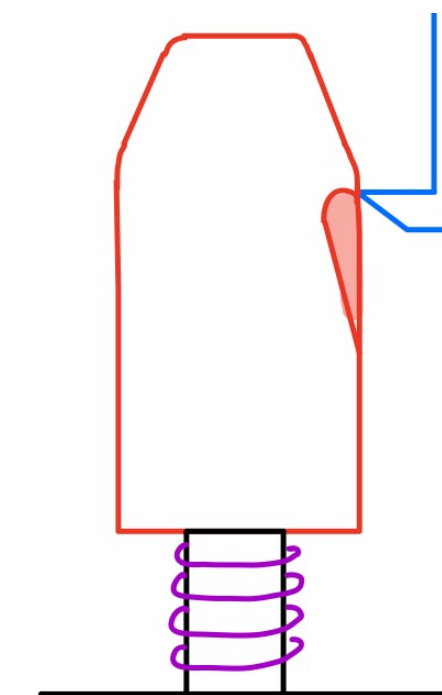


Retractable pens with rotating cam body

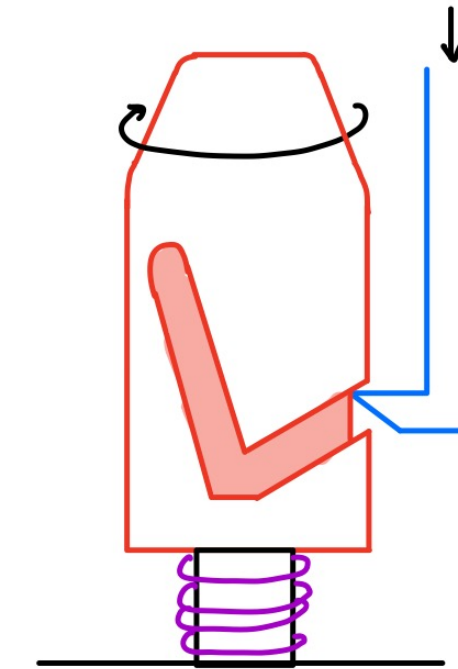
Concept



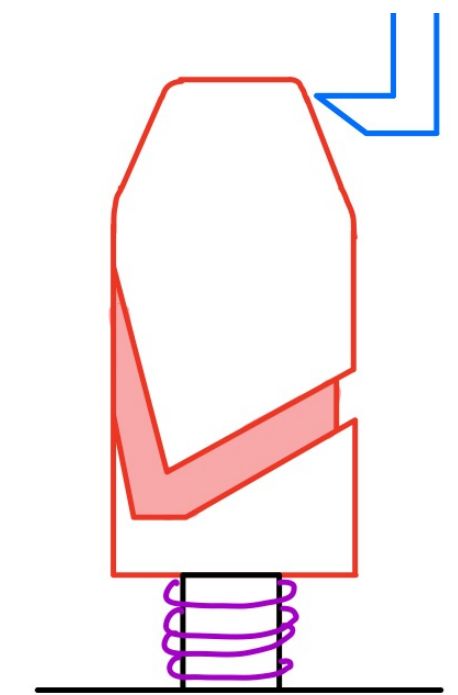
1. Hook sits in groove of rotating part when locked



3. Button is released and hook travels up path



2. Button is pressed and hook travels down path



4. Hook is removed and part rotates to initial position

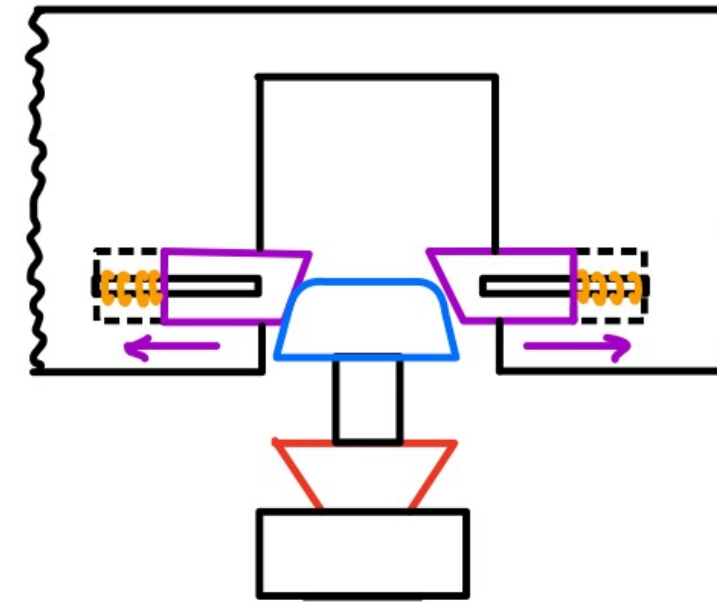
Sliding Latch Mechanism

Inspiration

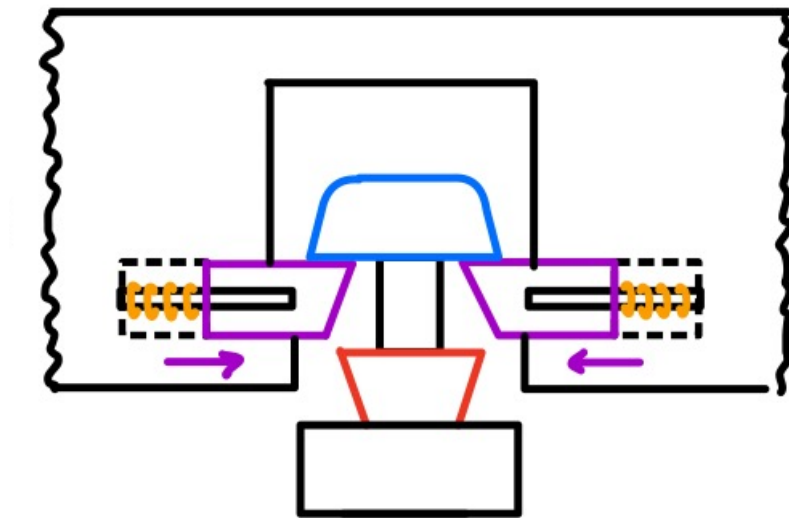


Knobs with strike plate and sliding latch

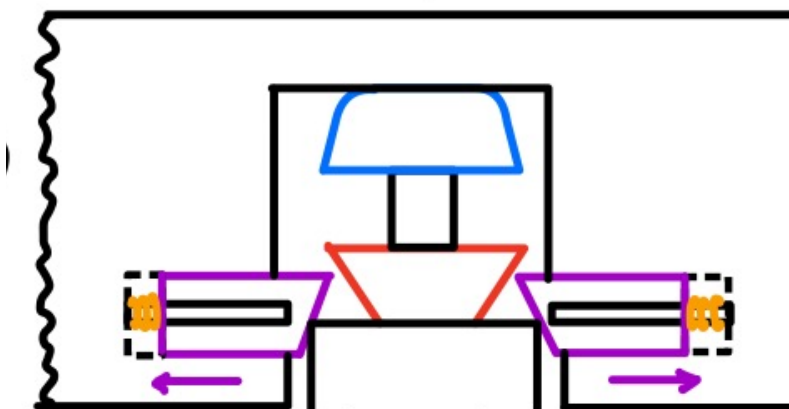
Concept



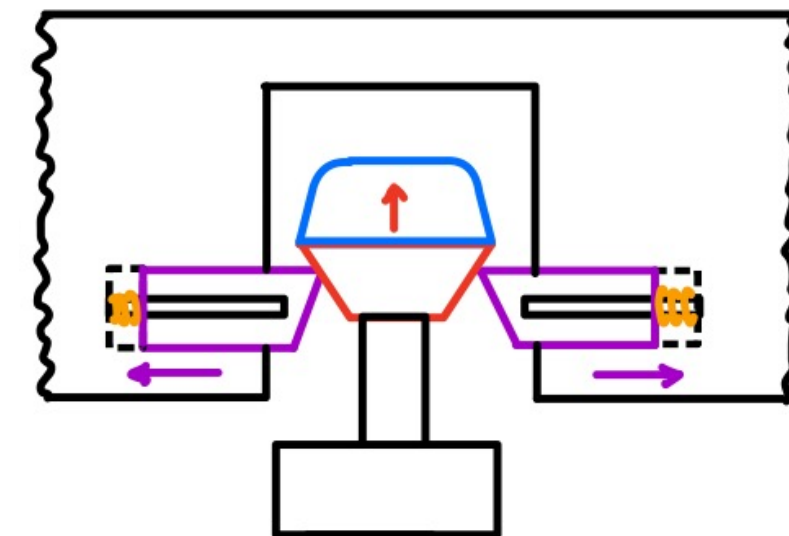
1. Button is pressed and latches slide over fixed part



2. Button is released and latches lock in place

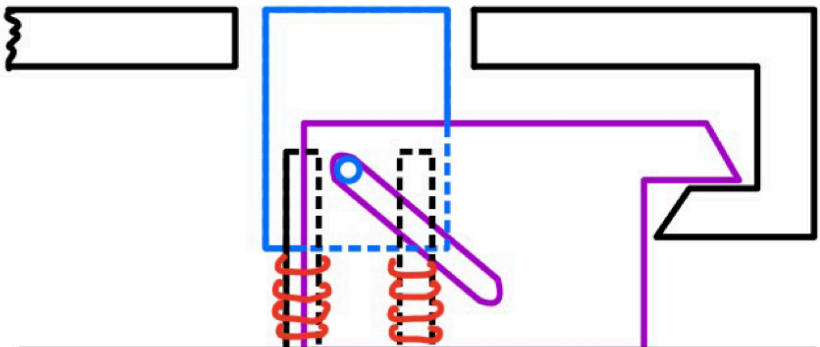
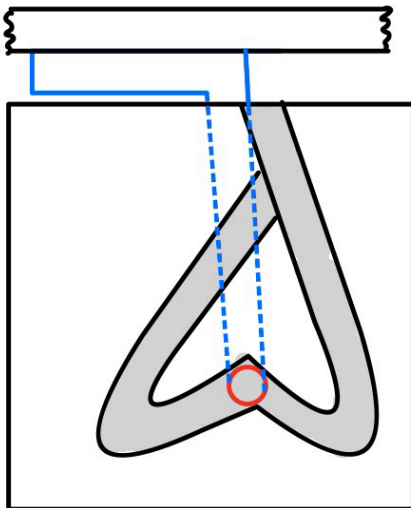
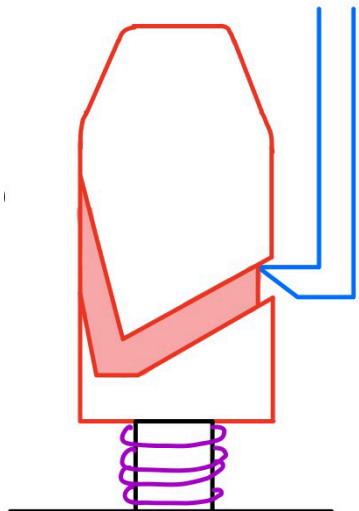
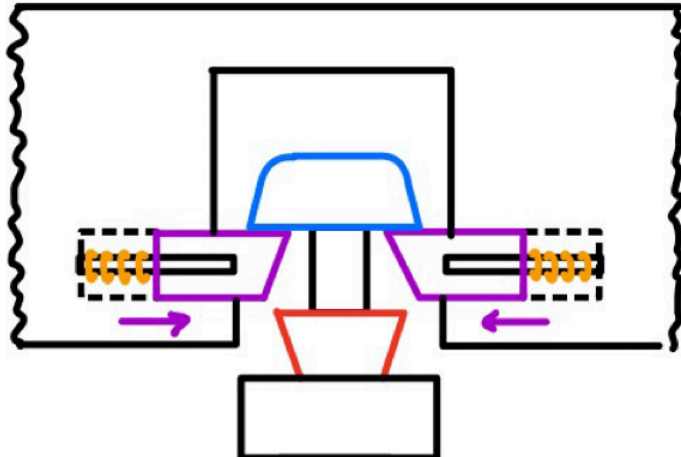


3. Button is pressed and latches slide over sliding part

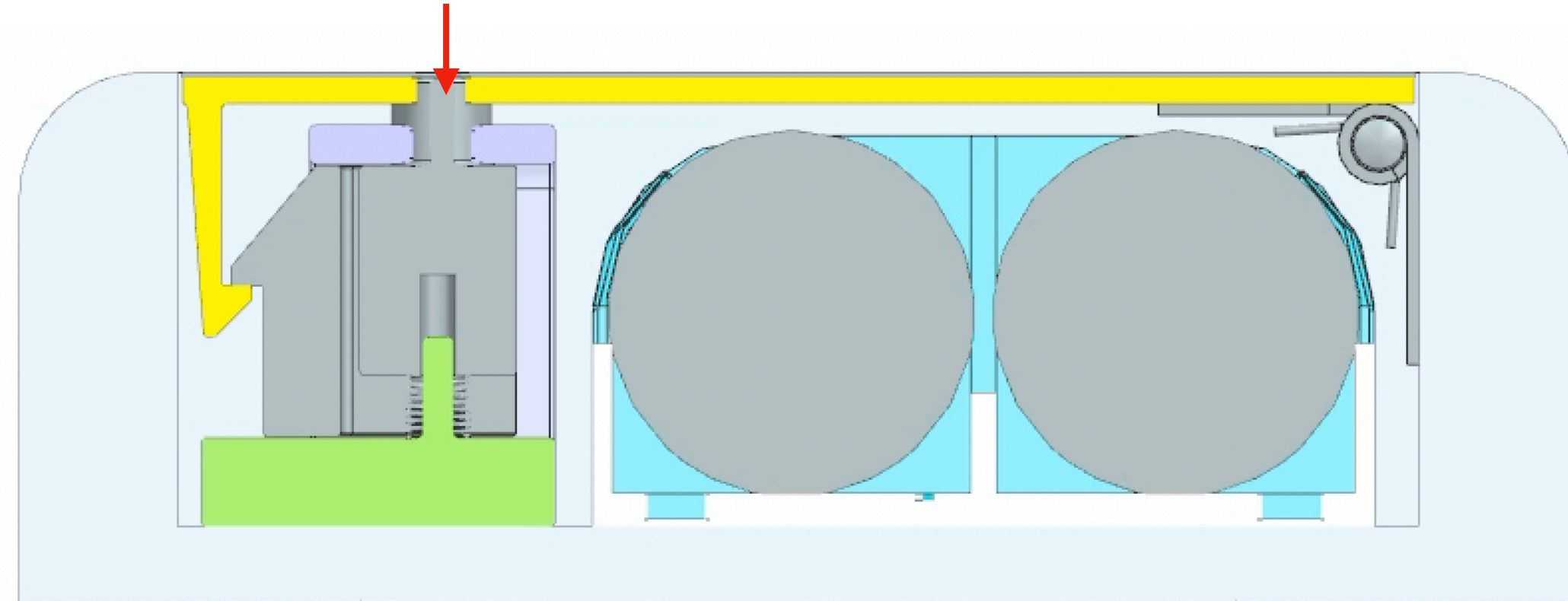


4. Button is released and latches to unlocked position

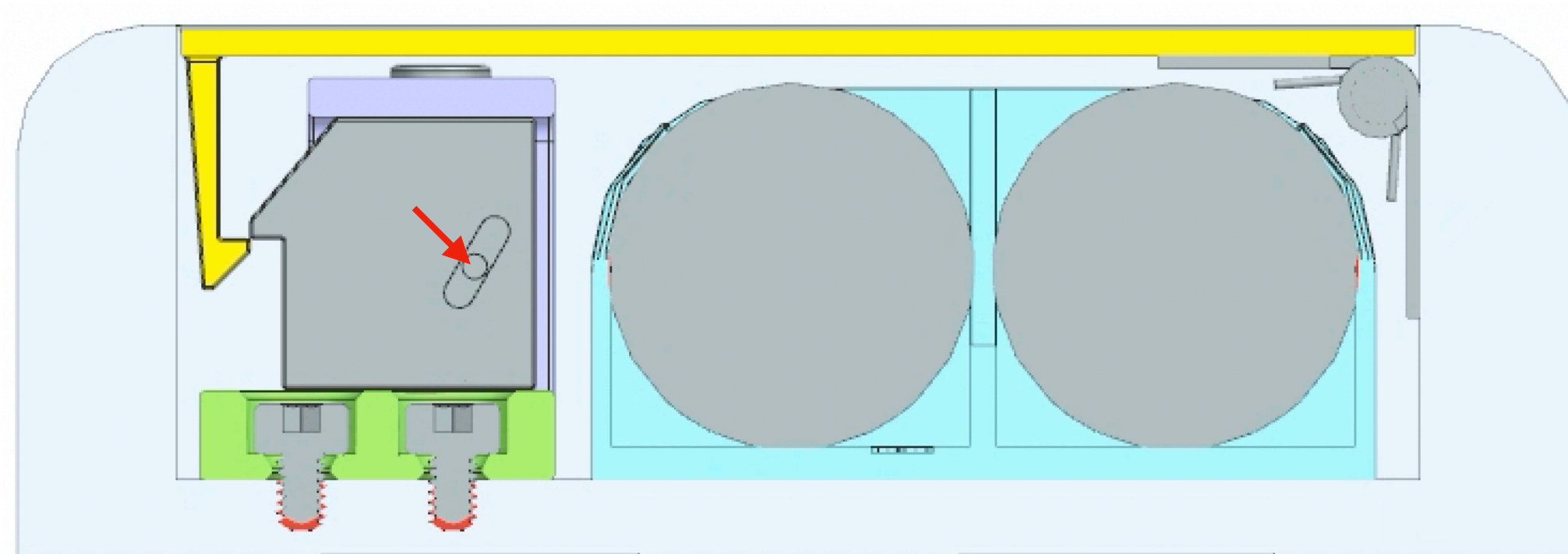
Comparing Mechanisms

Moving Snap Fit 	Push Release 	Rotating Body 	Sliding Latch 
+ Less sensitive to tolerances + Not easily actuated + Simple components + Less components	+ Button exists on door + Not easily actuated + Less components + Compact	+ Button exists on door + Not easily actuated + Less components	+ Button exists on door + Less sensitive to bending
- Sensitive to bending and wear - Button exists on housing (user must put finger through door)	- Groove may jam - Sensitive to bending in multiple directions - Sensitive to tolerances - Complex components	- Groove may jam - Sensitive to tolerances - Downward hook force may not trigger rotation - Requires more space - Complex components	- Sensitive to tolerances - User may push down too far during locking - More components - Requires more space - Complex components

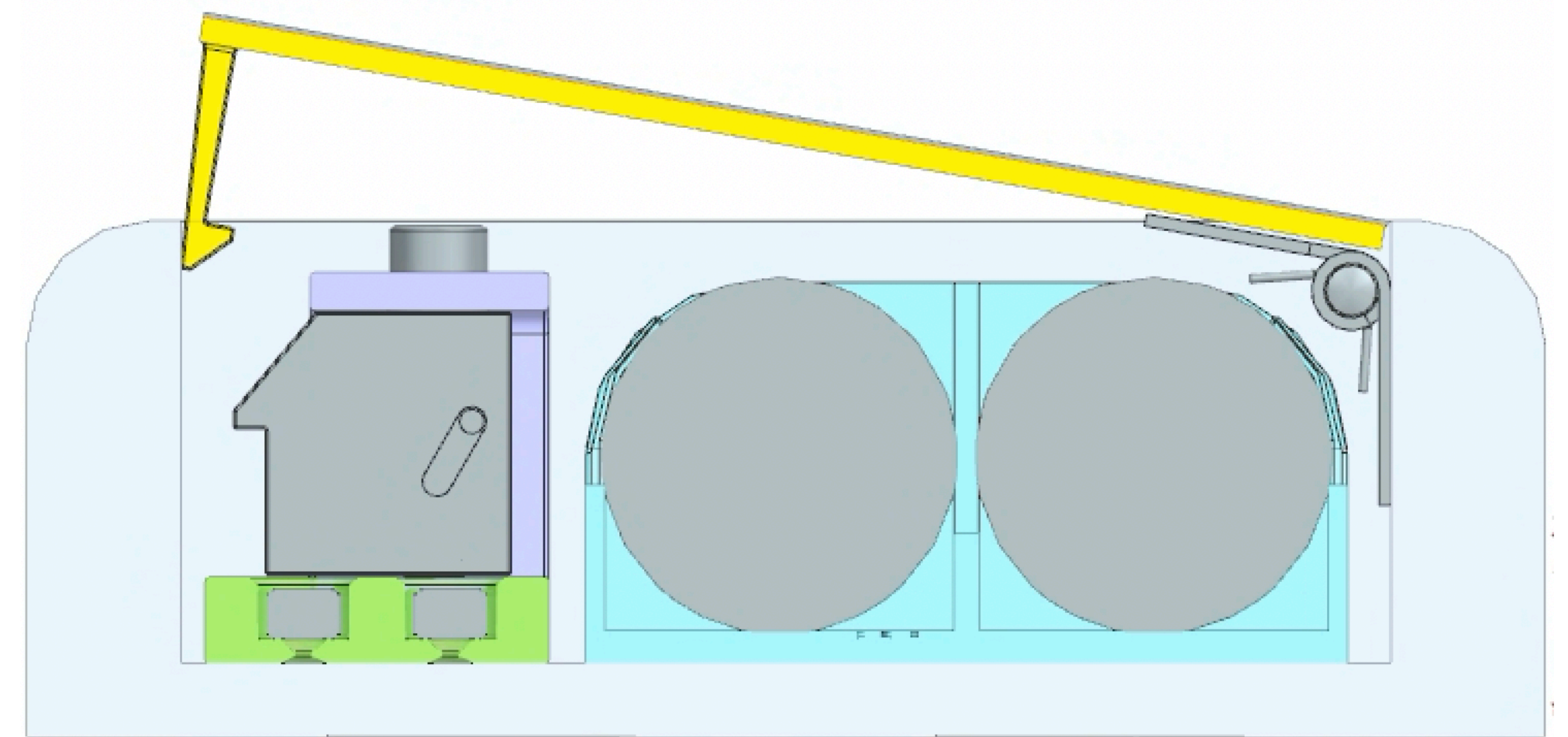
Sliding Snap Fit Design



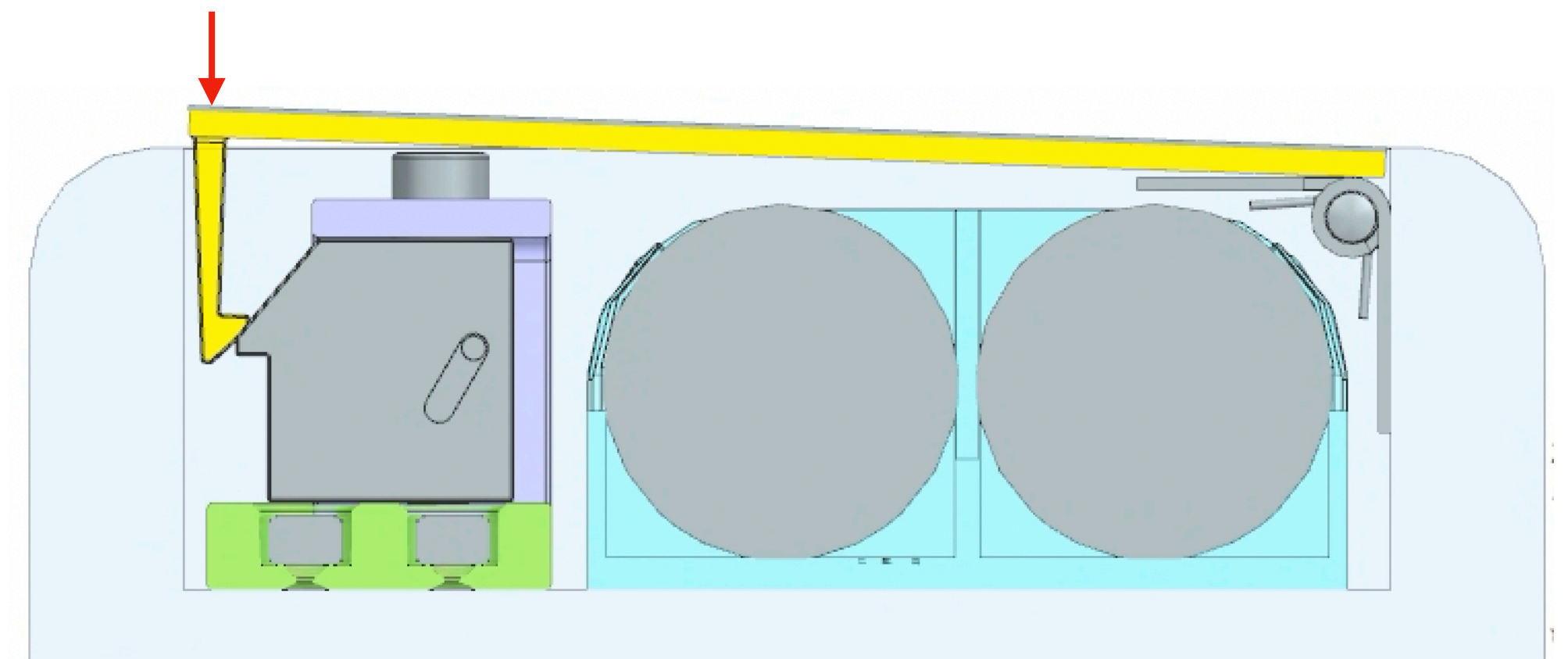
1. User presses down on the button, which is constrained by the chassis and pushed upward by the springs.



2. Button moves down and pegs apply a force to the latch slot, causing the latch to slide away from the door.



3. Downward force applied by latch to door is removed and spring loaded hinges cause door to pop open.



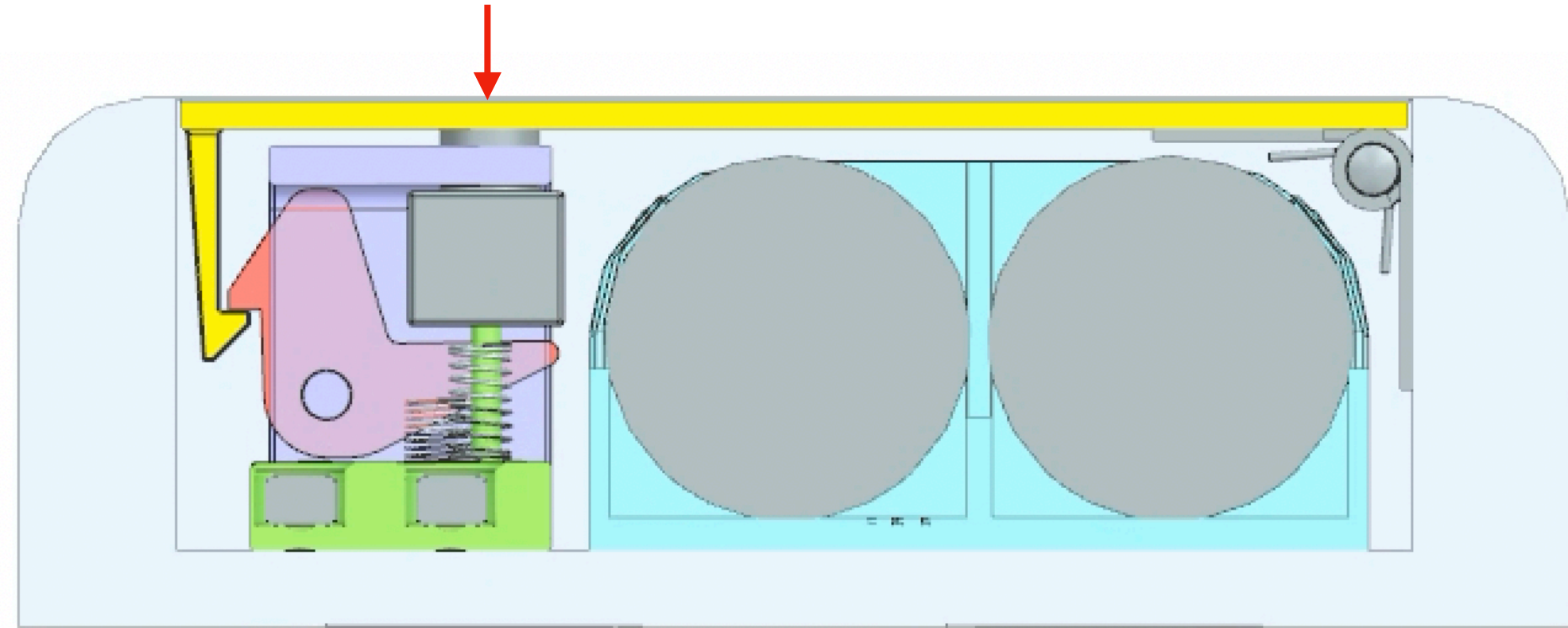
4. After replacing batteries, user applies a downward force to the door, engaging the snap fit mechanism.

Sliding Snap Fit Bill of Materials

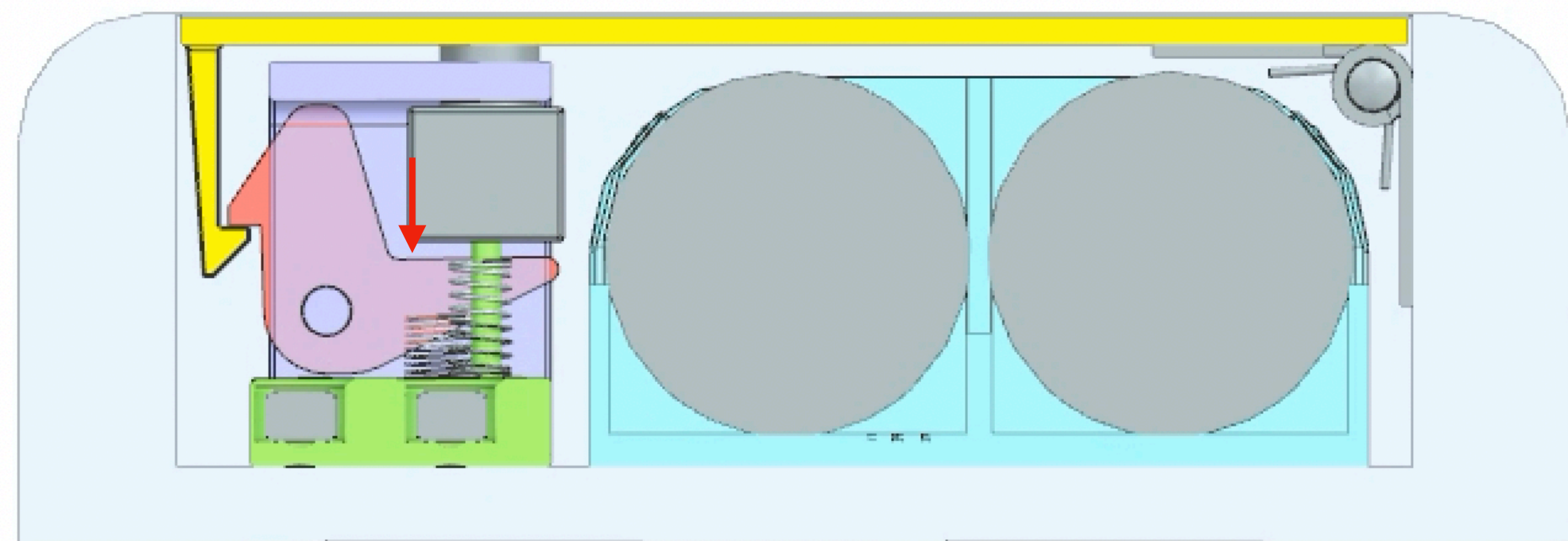
* Remaining components assumed out-of-scope

Component	Quantity	Material	Manufacturing	Mass (g)	Cost per Unit (\$)	Total Cost (\$)
Top Chassis	1	ABS	Injection Molding	6.9	\$0.35	\$0.35
Button	1	Aluminum 6061-T6	Die Casting	3.6	\$0.10	\$0.10
Latch	1	ABS	Injection Molding	4.6	\$0.25	\$0.25
Springs	2	<i>(Purchased)</i>	<i>(Purchased)</i>	<i>(Purchased)</i>	\$0.10	\$0.20
Screws	4	<i>(Purchased)</i>	<i>(Purchased)</i>	<i>(Purchased)</i>	\$0.10	\$0.40
Bottom Chassis	1	ABS	Injection Molding	8.6	\$0.45	\$0.45
Spring Loaded Hinge	2	<i>(Purchased)</i>	<i>(Purchased)</i>	<i>(Purchased)</i>	\$0.10	\$0.20
Door	1	ABS	Injection Molding	24.4	\$1.20	\$1.20
Door Cover	1	Aluminum 6061-T6	Stamping	4.7	\$0.15	\$0.15
					Total	\$3.30

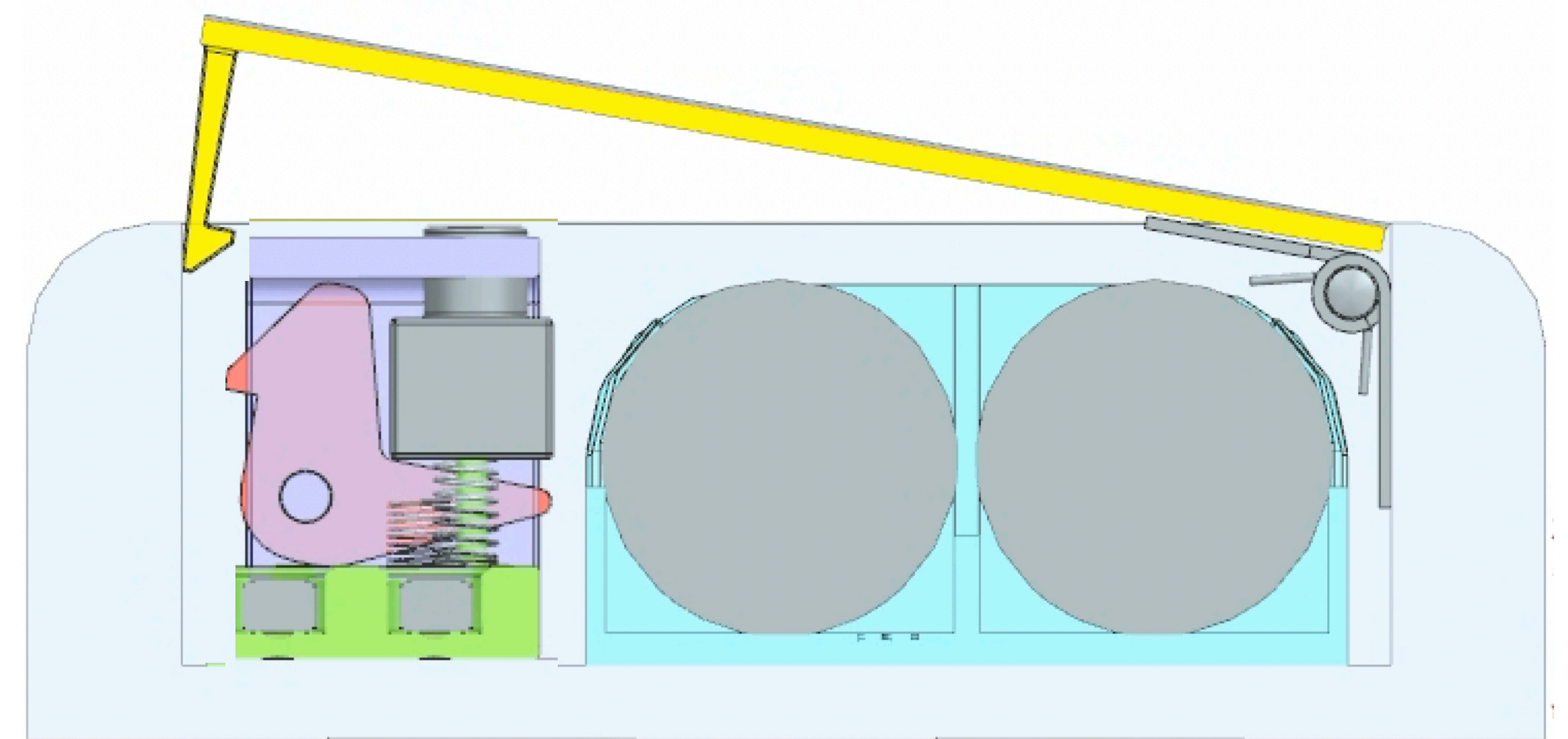
Rotating Snap Fit Design



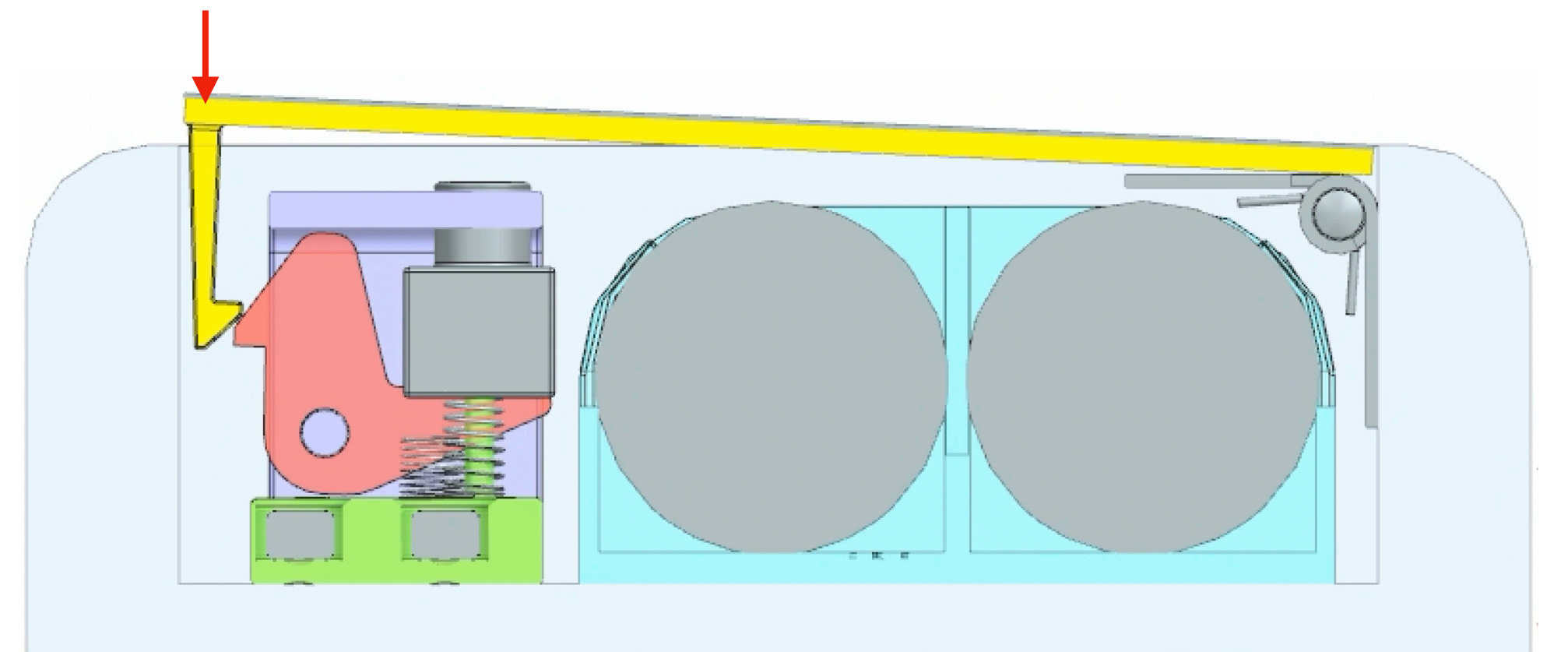
1. User presses down on the button, which is constrained by the top chassis and pushed upward by the springs.



2. Button moves down and applies a force to the tail of the latch, causing it to rotate about the peg.



3. Downward force applied by latch to door is removed and spring loaded hinges cause door to pop open.



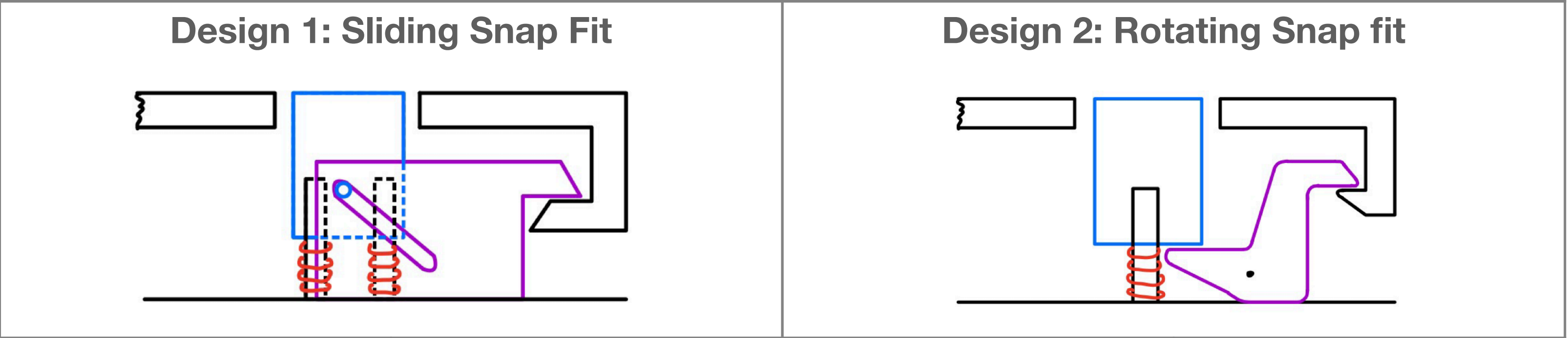
4. After replacing batteries, user applies a downward force to the door, engaging the snap fit mechanism.

Rotating Snap Fit Bill of Materials

* Remaining components assumed out-of-scope

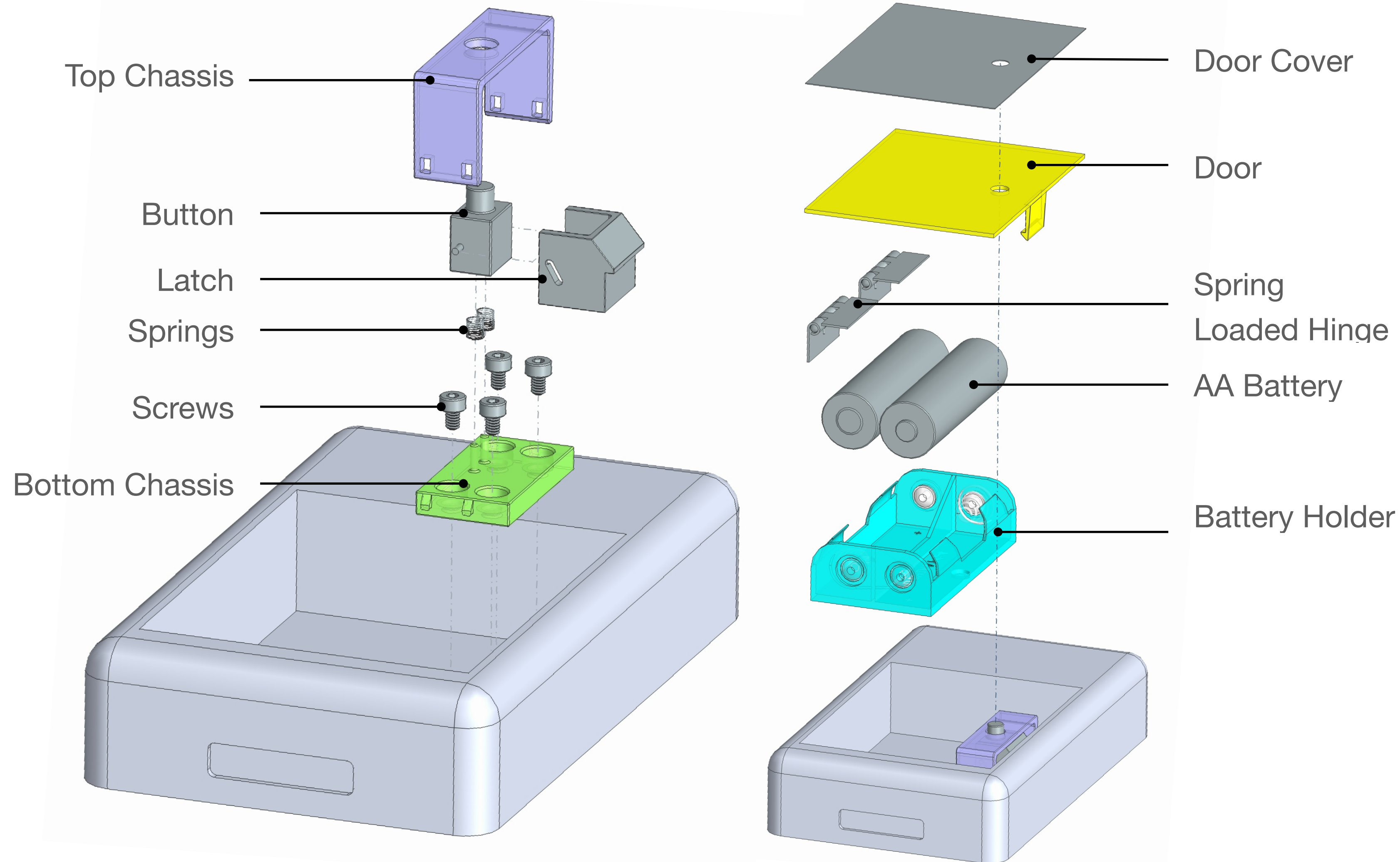
Component	Quantity	Material	Manufacturing	Mass (g)	Cost per Unit (\$)	Total Cost (\$)
Top Chassis	1	ABS	Injection Molding	8.5	\$0.45	\$0.45
Button	1	Aluminum 6061-T6	Die Casting	3.8	\$0.10	\$0.10
Latch	1	ABS	Injection Molding	2.4	\$0.15	\$0.15
Springs	2	(Purchased)	(Purchased)	(Purchased)	\$0.10	\$0.20
Screws	4	(Purchased)	(Purchased)	(Purchased)	\$0.10	\$0.40
Bottom Chassis	1	ABS	Injection Molding	7.3	\$0.35	\$0.35
Spring Loaded Hinge	2	(Purchased)	(Purchased)	(Purchased)	\$0.10	\$0.20
Door	1	ABS	Injection Molding	\$1.20	\$1.20	\$1.20
Door Cover	1	Aluminum 6061-T6	Stamping	\$0.15	\$0.15	\$0.15
					Total	\$3.20

Decision Matrix



Criterion	Weight	Rating	Weighted Score	Rating	Weighted Score
Effectiveness	5	4	20	3	15
User Experience	4	4	16	3	12
Manufacturability	4	3	12	3	12
Ease of Assembly	4	3	12	3	12
Compactness	4	3	12	4	16
Cost	3	3	9	4	12
Total Score			81		79

Sliding Snap Fit Components and Assembly

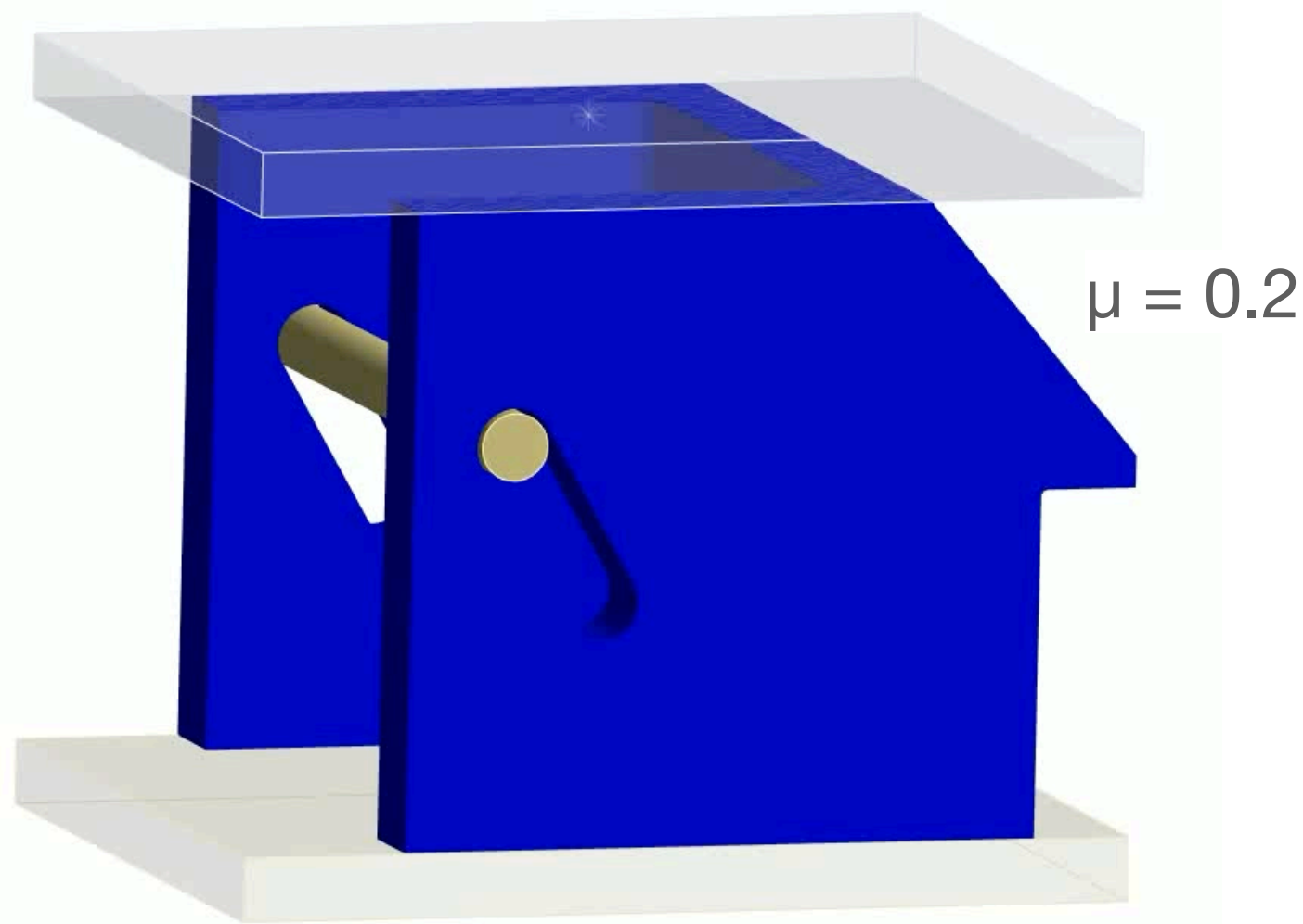


Procedure

1. Bottom chassis is fixed to housing via screws
2. Button is assembled into latch so that button pegs will slide along latch slots
3. Springs placed onto bottom chassis posts
4. Latch and button assembly positioned over bottom chassis posts
5. Top chassis secured to bottom chassis via snap fit features
6. Battery holder fixed to housing (assumed out-of-scope)
7. Door cover secured to door via adhesive
8. Hinges assembled to door and housing via adhesive or screws

Initial Button and Latch Calculations

FEA

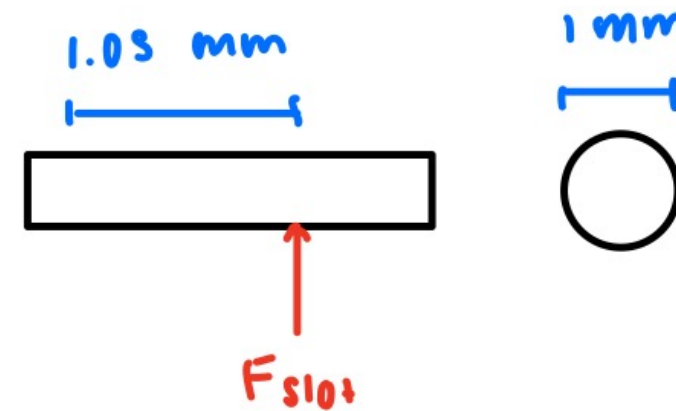


Force for button to slide latch: 0.0033 N

Assumptions

- Negligible friction between peg and slot
- Negligible friction between latch and door
- Rigid bodies: button peg moves with peg, snap fit moves with latch

Hand Calculations

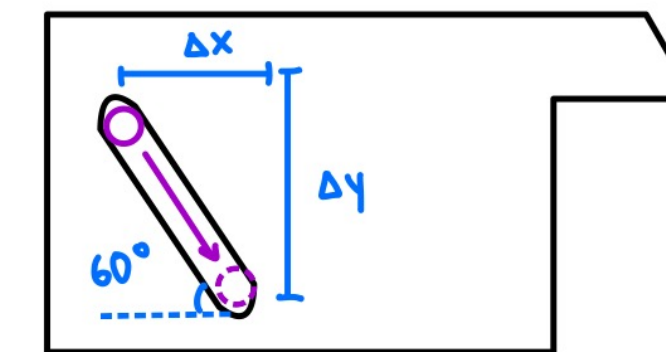


$$M = 1.05 F_{\text{slot}} = 1.05 \left(\frac{0.0033}{2} \right) = 0.0017 \text{ N} \cdot \text{mm}$$

$$I = \frac{\pi}{4} r^4 = \frac{\pi}{4} (0.5)^4 = 0.05 \text{ mm}^4$$

$$\sigma_b = \frac{My}{I} = \frac{(0.0017)(0.5)}{0.05} = 0.0173 \text{ MPa}$$

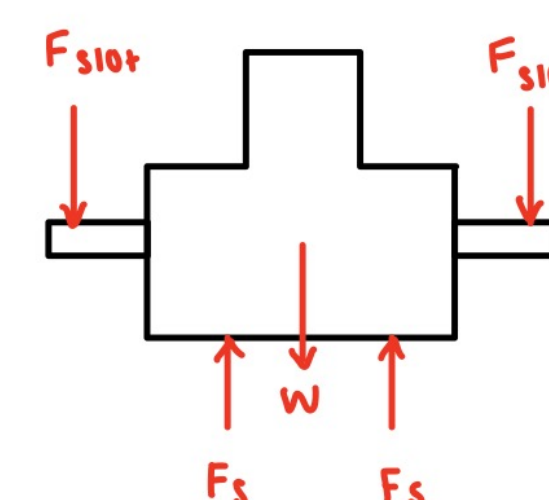
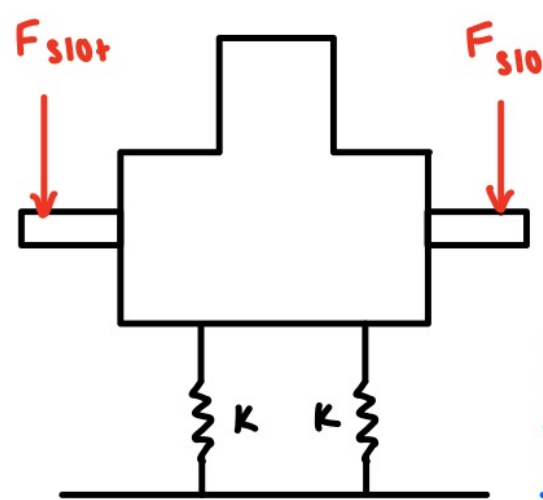
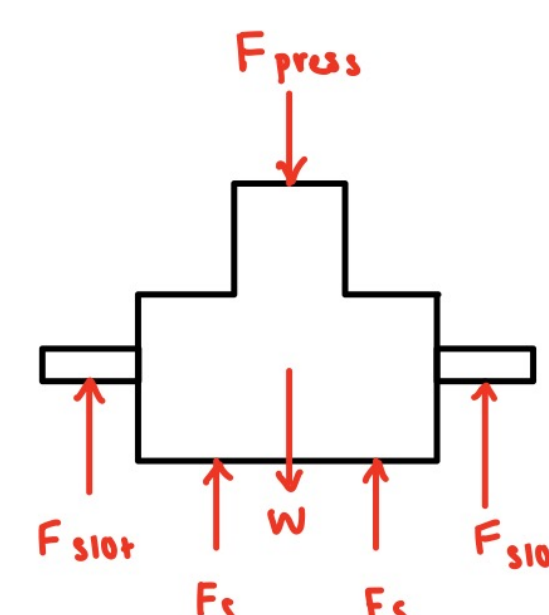
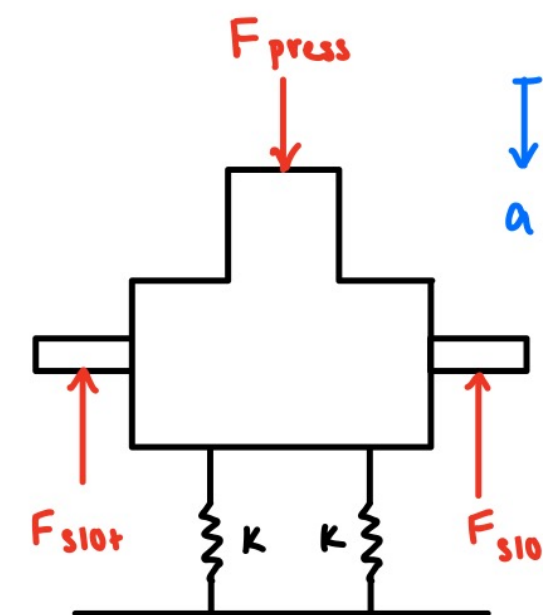
Max peg bending stress < fatigue strength of SUS316 for 1000 cycles (300 MPa)



$$\tan(60^\circ) = \frac{\Delta y}{0.8 \text{ mm}}$$

$$\Delta y = 1.39 \text{ mm}$$

Button travel: 1.39 mm



Required spring constant: 0.01-0.875 N/mm

$$m = 0.0033 \text{ Kg} \quad F_s = K \Delta y = 1.73 K$$

$$\sum F = F_{\text{press}} + W - 2F_{y,\text{slot}} - 2F_s = ma$$

$$3 + 0.0324 - 0.0033 - 2(1.73)K > 0$$

$$3.0291 > 3.46K \rightarrow K < 0.875$$

$$W = mg = (0.0033)(9.81) = 0.0324 \text{ N}$$

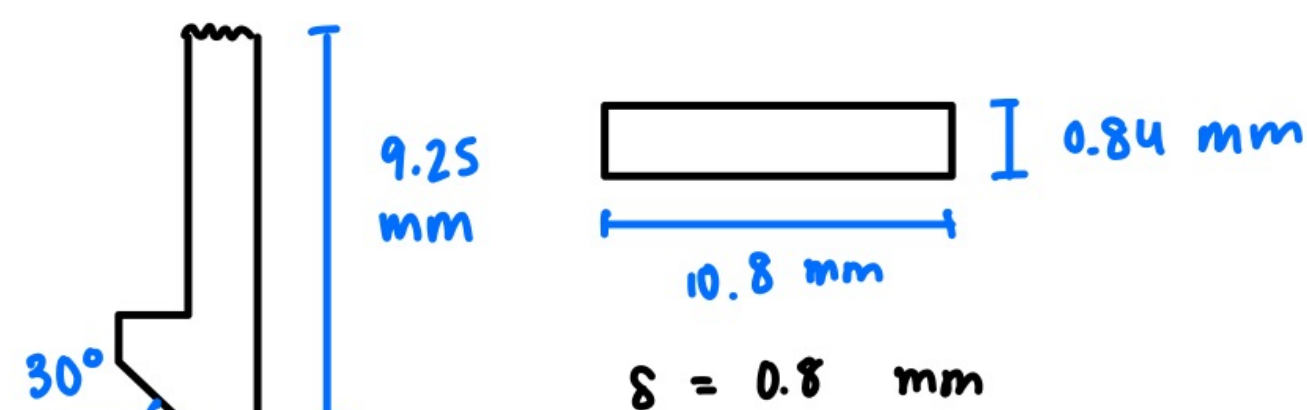
$$\sum F = 2F_s - 2F_{\text{slot}} - W = ma$$

$$2(1.73)K - 0.0033 - 0.0324 > 0$$

$$3.46K > 0.0357 \rightarrow K > 0.01$$

Initial Door and Latch Calculations

Hand Calculations



$$\epsilon_{max} = 1.5 \frac{\epsilon \delta}{L^2 Q} = 1.5 \frac{(0.84)(0.8)}{9.25^2} = 0.012$$

Max strain < allowable strain for ABS (7%)

$$E = 2 \text{ GPa} = 2000 \text{ MPa}$$

$$\sigma_{max} = E \epsilon_{max} = (2000)(0.012) = 24 \text{ MPa}$$

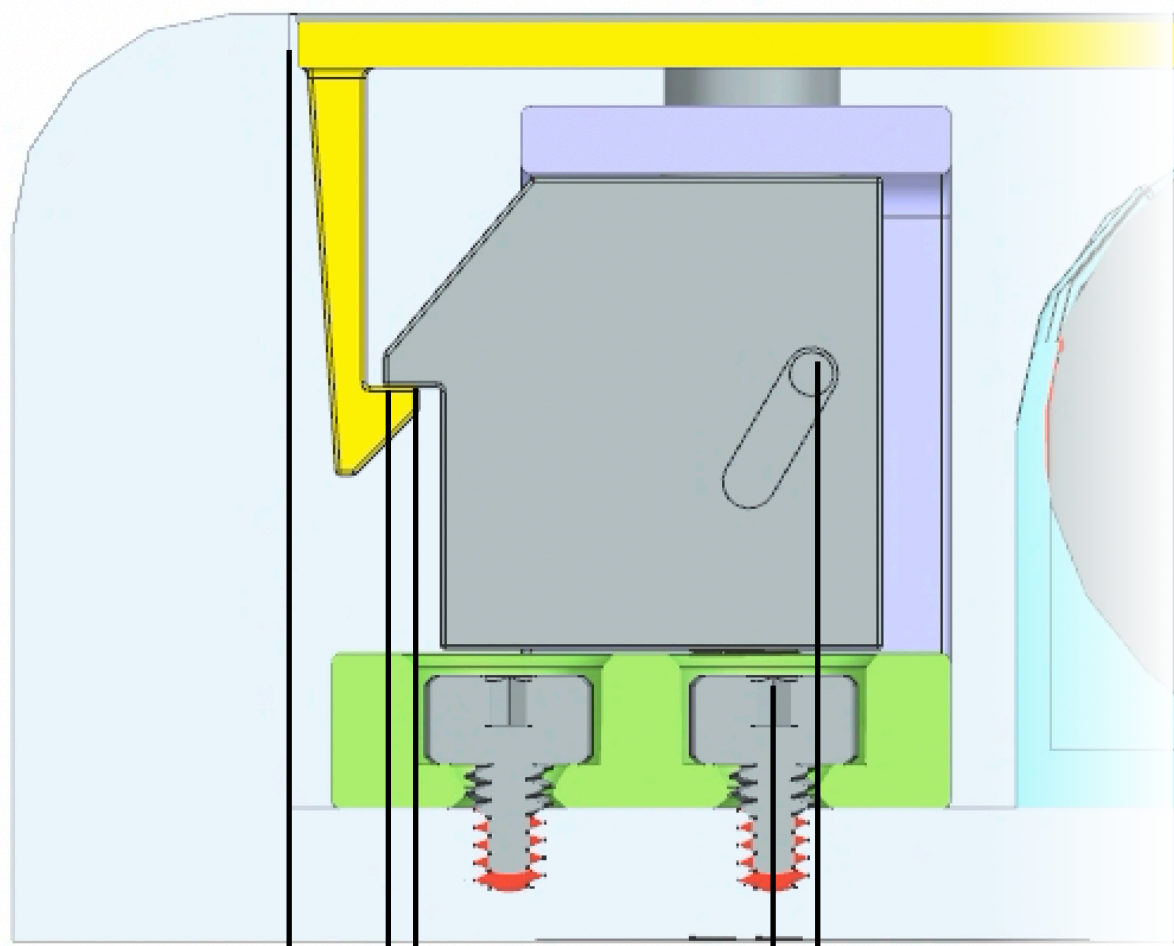
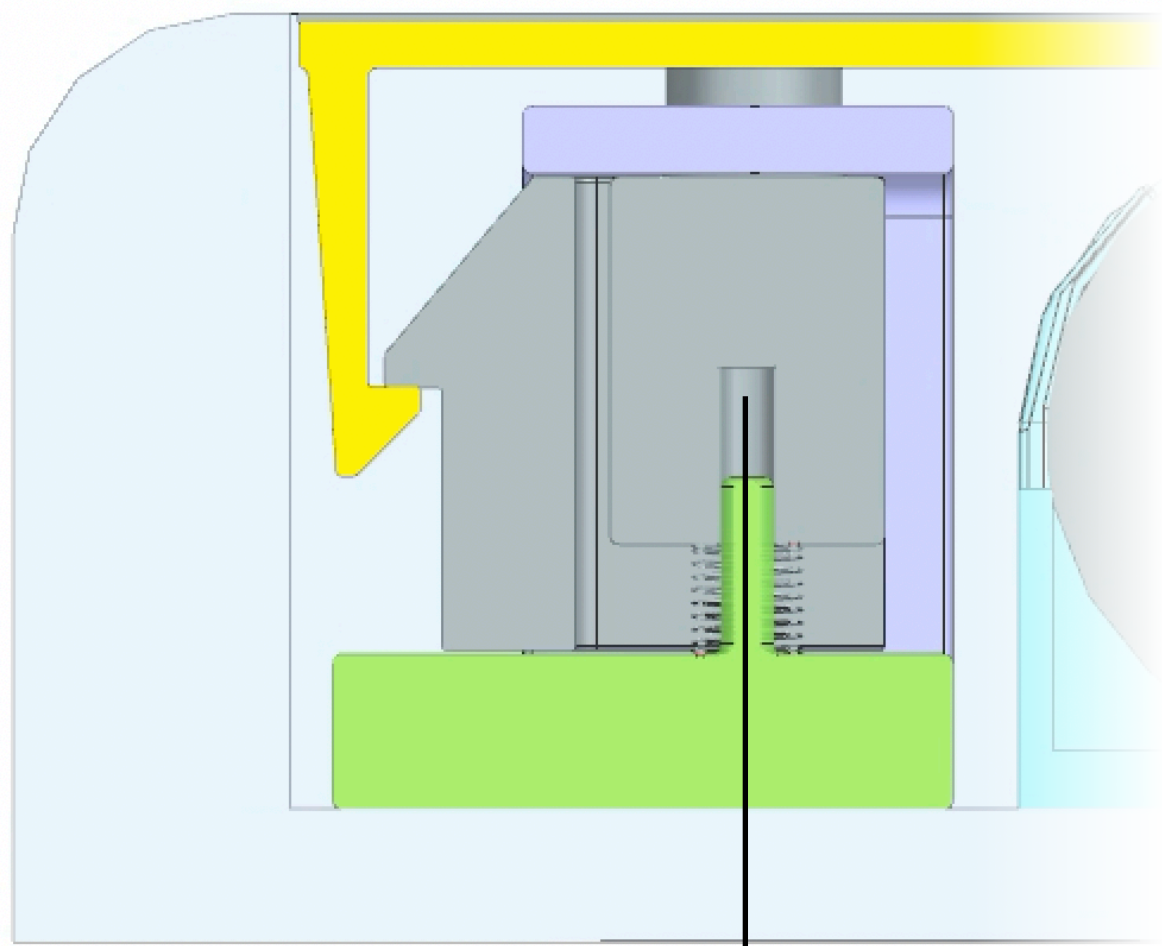
Max door stress < fatigue strength of ABS for 1000 cycles (35 MPa)

$$P = \frac{bt^2 \epsilon E}{6L} = \frac{(10.8)(0.84)^2 (2000)(0.012)}{6(9.25)} = 3.3 \text{ N}$$

$$W = P \frac{\mu + \tan \alpha}{1 - \mu \tan \alpha} = 3.3 \frac{0.2 + \tan 50^\circ}{1 - 0.2 \tan 50^\circ} = 6.1 \text{ N}$$

Required force to close door: 6.1 N

Tolerance Analysis



No.	Description	Nom.	+ Tol.	- Tol.
1 - 2	Latch tip to latch slot center	9.70	0.05	0.05
2 - 3	Latch slot center to button peg center	0.00	0.10	0.00
3 - 4	Button peg center to button hole	-1.50	0.05	0.05
4 - 5	Button hole to chassis post	0.00	0.10	0.10
5 - 6	Chassis post to screw hole center	0.64	0.05	0.05
4 - 5	Chassis screw hole center to housing housing screw hole center	0.00	0.10	0.10
5 - 6	Housing screw hole center to edge	-11.00	0.10	0.10
6 - 7	Housing edge to door tip	2.95	0.20	0.20

Nominal overlap:
0.79 mm

RSS Max overlap:
1.09 mm

RSS Min overlap:
0.51 mm